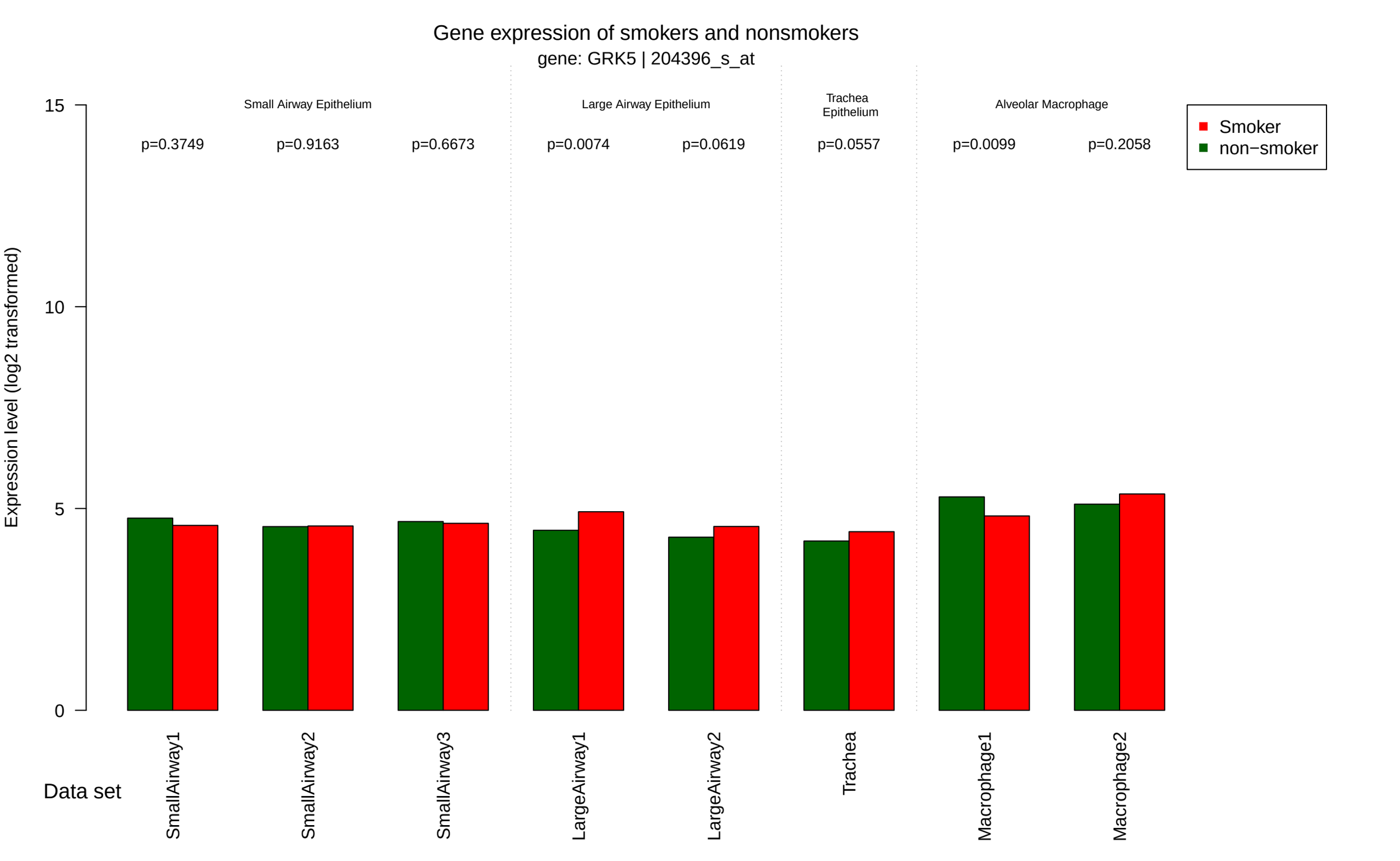
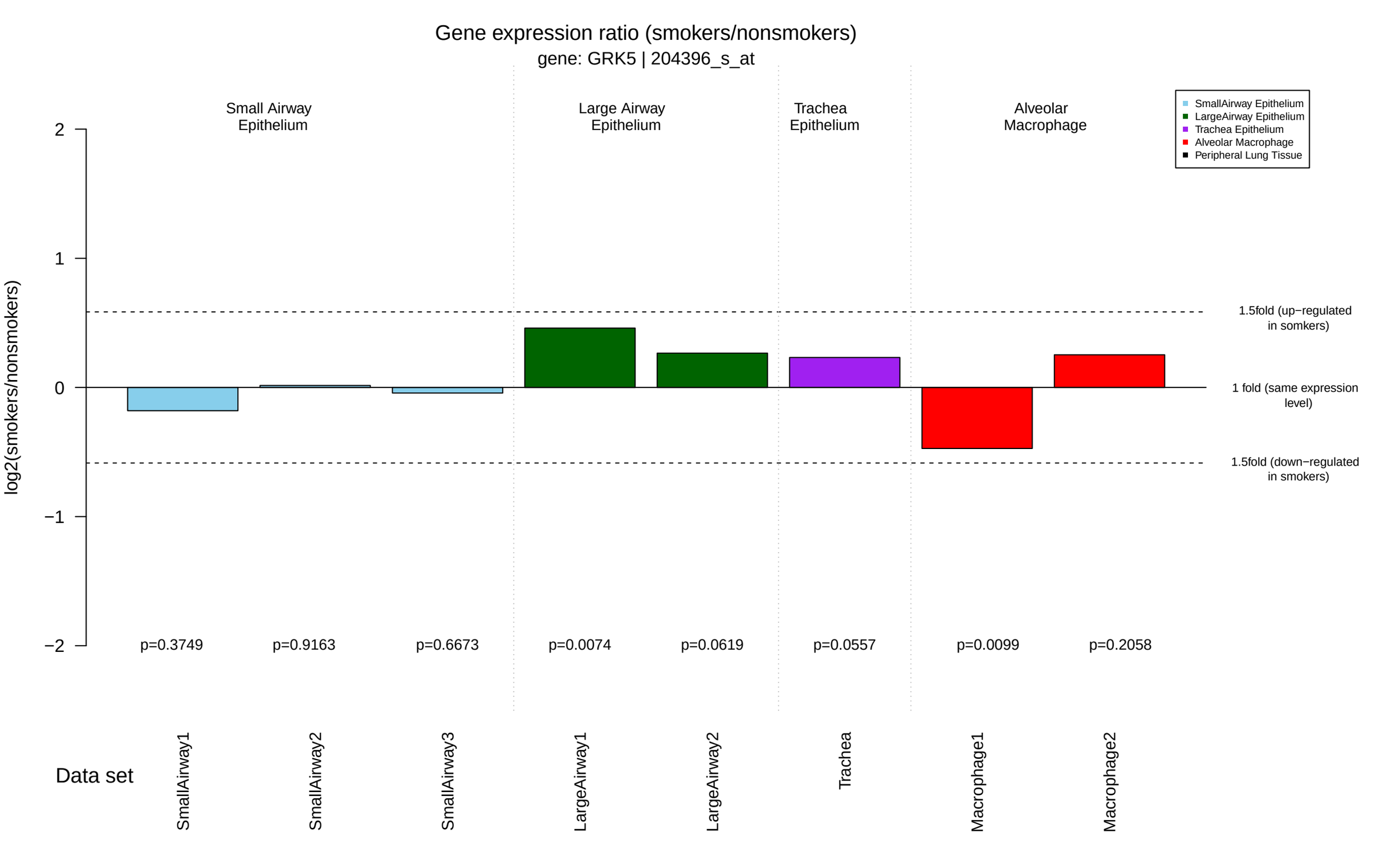
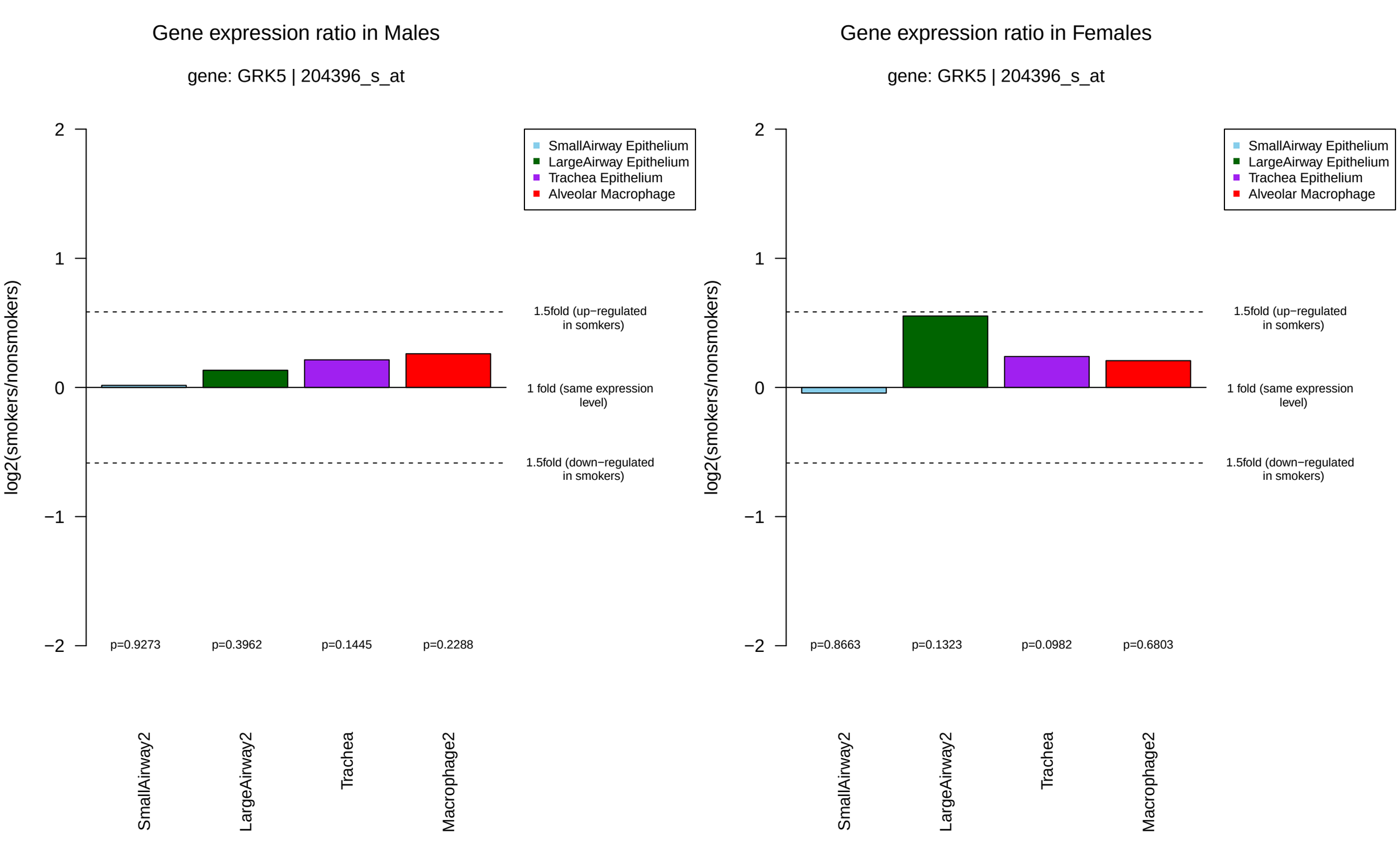
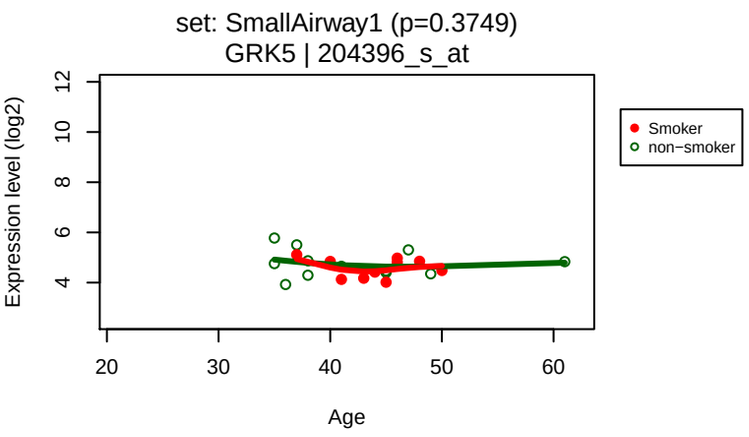
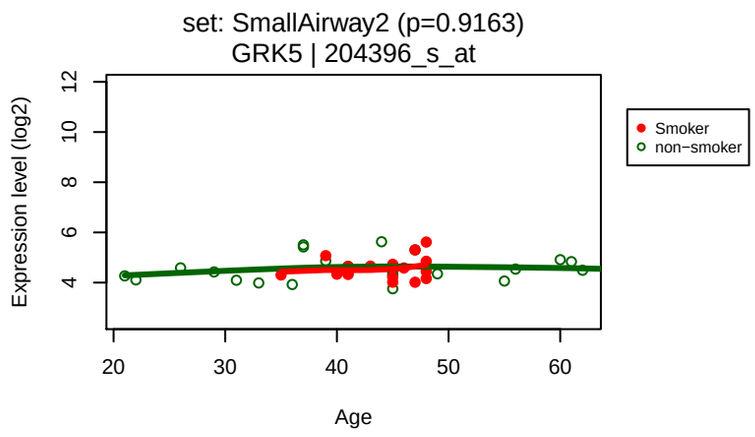
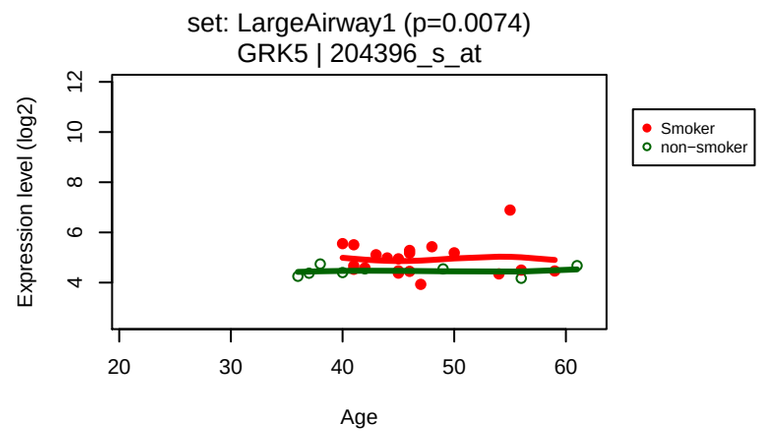

Gene expression of smokers and nonsmokers
gene: GRK5 | 204396_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3749
p=0.9163
p=0.6673
p=0.0074
p=0.0619
p=0.0557
p=0.0099
p=0.2058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

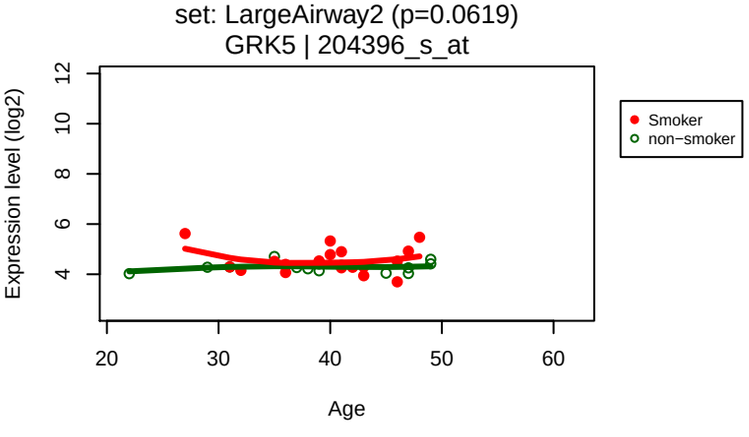

Gene expression ratio (smokers/nonsmokers)
gene: GRK5 | 204396_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3749
p=0.9163
p=0.6673
p=0.0074
p=0.0619
p=0.0557
p=0.0099
p=0.2058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

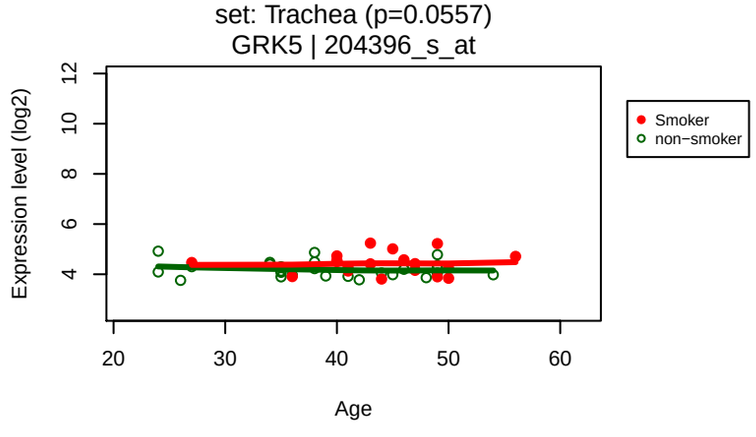

Gene expression ratio in Males
gene: GRK5 | 204396_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9273
p=0.3962
p=0.1445
p=0.2288
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK5 | 204396_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8663
p=0.1323
p=0.0982
p=0.6803
SmallAirway2
LargeAirway2
Trachea
Macrophage2

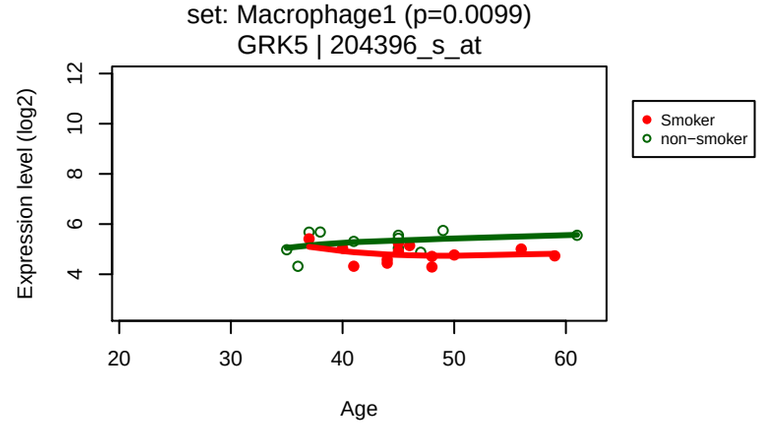

set: SmallAirway1 (p=0.3749)
GRK5 | 204396_s_at
Expression level (log2)
Age
Smoker
non−smoker

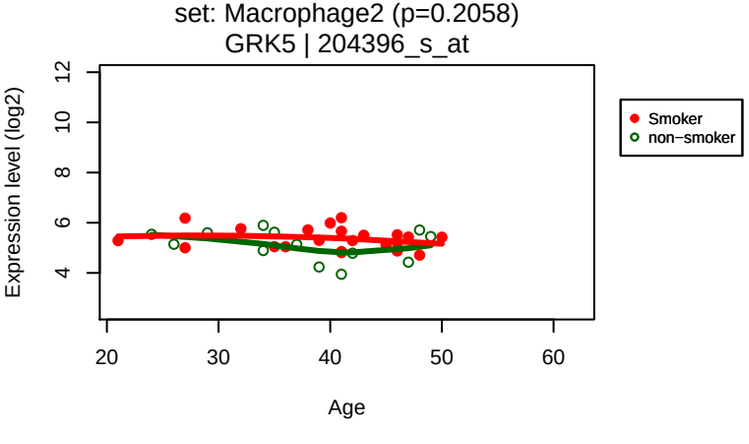

set: SmallAirway2 (p=0.9163)
GRK5 | 204396_s_at
Expression level (log2)
Age
Smoker
non−smoker

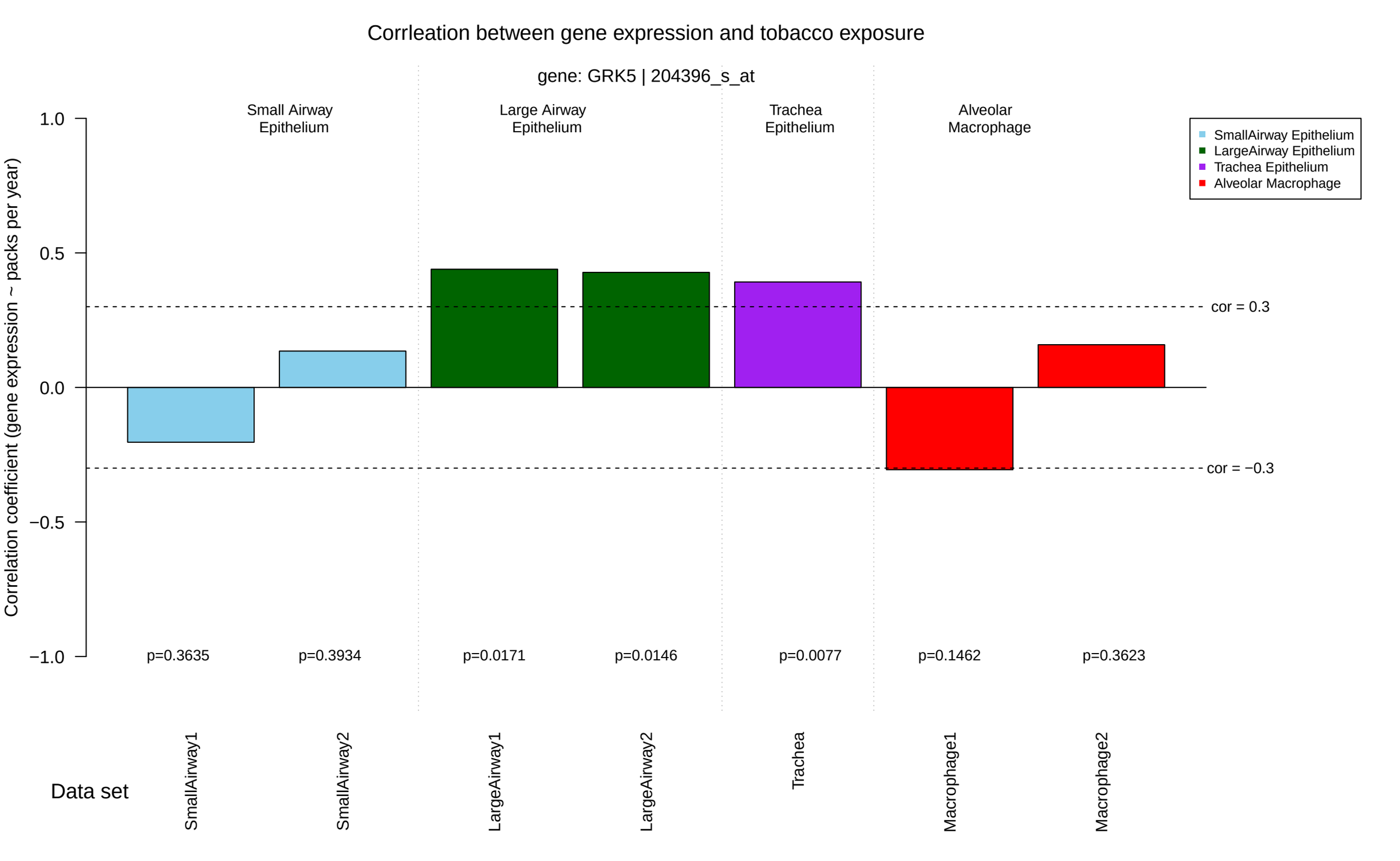

set: LargeAirway1 (p=0.0074)
GRK5 | 204396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0619)
GRK5 | 204396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0557)
GRK5 | 204396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0099)
GRK5 | 204396_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2058)
GRK5 | 204396_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK5 | 204396_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3635
p=0.3934
p=0.0171
p=0.0146
p=0.0077
p=0.1462
p=0.3623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2